

Practical 2

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Aim - Implement Gradient descent Find Correlation between Math and CS column

1. Find value of m (slope) and b (intercept).

2. Compare the cost between each iteration and stop when it reaches a certain threshold value. (To compare the threshold use `isClose()` of numpy library) <https://numpy.org/doc/stable/reference/generated/numpy.isclose.html>
(<https://numpy.org/doc/stable/reference/generated/numpy.isclose.html>)

3. You have to print how many iterations are required to figure out the value of m (slope) and b (intercept)

In [2]:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

In [3]:

```
df = pd.read_csv("test_scores.csv")
df
```

Out[3]:

	name	math	cs
0	david	92	98
1	laura	56	68
2	sanjay	88	81
3	wei	70	80
4	jeff	80	83
5	aamir	49	52
6	venkat	65	66
7	virat	35	30
8	arthur	66	68
9	paul	67	73

In [4]:

```
%matplotlib inline
def gradient_descent(x,y):
    m_curr = b_curr = 0
    iteration=100
    rate = 0.008
    n = len(x)
    temp =0
    #plt.scatter(x,y,color='red',marker='+',linewidth='5')
    for i in range(iteration):
        y_predicted = m_curr * x + b_curr
        print (m_curr,b_curr, i)
        cost=(1/n) * sum({val**2 for val in (y-y_predicted)})
        md = -(2/n)*sum(x*(y-y_predicted))
        bd = -(2/n)*sum(y-y_predicted)
        m_curr = m_curr - rate * md      #m=m-learning rate * d/dm
        b_curr = b_curr - rate * bd      #b=b-learning rate * d/db
        print("m-->{}, b-->{}, iteration--> {}".format(m_curr,b_curr,i))
        if np.isclose(cost,temp):
            print("Yes is close")
        temp = cost
```

In [5]:

```
X=np.array(df.math)
y=np.array(df.cs)
```

In [6]:

```
gradient_descent(X,y)
```

```
0 0 0
m-->79.13440000000001, b-->1.1184, iteration--> 0
79.13440000000001 1.1184 1
m-->-5836.882539520004, b-->-82.35994112000002, iteration--> 1
-5836.882539520004 -82.35994112000002 2
m-->436439.1092389523, b-->6158.5362761769, iteration--> 2
436439.1092389523 6158.5362761769 3
m-->-32627707.890128233, b-->-460405.0018588341, iteration--> 3
-32627707.890128233 -460405.0018588341 4
m-->2439217558.540131, b-->34419456.78953996, iteration--> 4
2439217558.540131 34419456.78953996 5
m-->-182353664631.80585, b-->-2573166979.968385, iteration--> 5
-182353664631.80585 -2573166979.968385 6
m-->13632592510187.02, b-->192367600451.30362, iteration--> 6
13632592510187.02 192367600451.30362 7
m-->-1019160097072701.9, b-->-14381225156042.686, iteration--> 7
-1019160097072701.9 -14381225156042.686 8
m-->7.61914729490489e+16, b-->1075127186197759.0, iteration--> 8
7.61914729490489e+16 1075127186197759.0 9
m-->-5.696004550040318e+18, b-->-8.037552113672486e+16, iteration--> 9
-5.696004550040318e+18 -8.037552113672486e+16 10
m-->4.258280694452045e+20, b-->6.008800150284555e+18, iteration--> 10
4.258280694452045e+20 6.008800150284555e+18 11
m-->-3.183451542821299e+22, b-->-4.492123812751546e+20, iteration--> 11
-3.183451542821299e+22 -4.492123812751546e+20 12
m-->2.3799191393594588e+24, b-->3.35827051064993e+22, iteration--> 12
2.3799191393594588e+24 3.35827051064993e+22 13
m-->-1.7792056934750122e+26, b-->-2.510612194322595e+24, iteration--> 13
-1.7792056934750122e+26 -2.510612194322595e+24 14
m-->1.3301178377622923e+28, b-->1.8769106211939587e+26, iteration--> 14
1.3301178377622923e+28 1.8769106211939587e+26 15
m-->-9.94383880864241e+29, b-->-1.4031611444877895e+28, iteration--> 15
-9.94383880864241e+29 -1.4031611444877895e+28 16
m-->7.433922577762912e+31, b-->1.0489903862059409e+30, iteration--> 16
7.433922577762912e+31 1.0489903862059409e+30 17
m-->-5.557532252447888e+33, b-->-7.842155797110337e+31, iteration--> 17
-5.557532252447888e+33 -7.842155797110337e+31 18
m-->4.154760076381245e+35, b-->5.862723658372736e+33, iteration--> 18
4.154760076381245e+35 5.862723658372736e+33 19
m-->-3.1060604793950054e+37, b-->-4.382918368837888e+35, iteration--> 19
-3.1060604793950054e+37 -4.382918368837888e+35 20
m-->2.3220622910342642e+39, b-->3.2766295236280173e+37, iteration--> 20
2.3220622910342642e+39 3.2766295236280173e+37 21
m-->-1.7359524449741356e+41, b-->-2.4495781421449225e+39, iteration--> 21
-1.7359524449741356e+41 -2.4495781421449225e+39 22
m-->1.2977821063833003e+43, b-->1.8312821242696497e+41, iteration--> 22
1.2977821063833003e+43 1.8312821242696497e+41 23
m-->-9.702099850285761e+44, b-->-1.369049699199658e+43, iteration--> 23
-9.702099850285761e+44 -1.369049699199658e+43 24
m-->7.253200752416095e+46, b-->1.0234889829584176e+45, iteration--> 24
7.253200752416095e+46 1.0234889829584176e+45 25
m-->-5.422426275410872e+48, b-->-7.651509648259214e+46, iteration--> 25
-5.422426275410872e+48 -7.651509648259214e+46 26
m-->4.053756088644308e+50, b-->5.72019834822027e+48, iteration--> 26
4.053756088644308e+50 5.72019834822027e+48 27
m-->-3.030550825695759e+52, b-->-4.276367755796549e+50, iteration--> 27
```

```
-3.030550825695759e+52 -4.276367755796549e+50 28
m-->2.2656119673437777e+54, b-->3.196973263786589e+52, iteration--> 28
2.2656119673437777e+54 3.196973263786589e+52 29
m-->-1.6937507013738668e+56, b-->-2.3900278537813695e+54, iteration--> 29
-1.6937507013738668e+56 -2.3900278537813695e+54 30
m-->1.2662324704118953e+58, b-->1.7867628755471802e+56, iteration--> 30
1.2662324704118953e+58 1.7867628755471802e+56 31
m-->-9.466237669008056e+59, b-->-1.3357675176808497e+58, iteration--> 31
-9.466237669008056e+59 -1.3357675176808497e+58 32
m-->7.076872351646279e+61, b-->9.986075296896015e+59, iteration--> 32
7.076872351646279e+61 9.986075296896015e+59 33
m-->-5.290604782242226e+63, b-->-7.465498188518088e+61, iteration--> 33
-5.290604782242226e+63 -7.465498188518088e+61 34
m-->3.9552075508855186e+65, b-->5.581137889085473e+63, iteration--> 34
3.9552075508855186e+65 5.581137889085473e+63 35
m-->-2.956876843851458e+67, b-->-4.1724074335578424e+65, iteration--> 35
-2.956876843851458e+67 -4.1724074335578424e+65 36
m-->2.2105339750748323e+69, b-->3.119253481562229e+67, iteration--> 36
2.2105339750748323e+69 3.119253481562229e+67 37
m-->-1.6525749001420424e+71, b-->-2.3319252583014086e+69, iteration--> 37
-1.6525749001420424e+71 -2.3319252583014086e+69 38
m-->1.2354498195337755e+73, b-->1.743325908730129e+71, iteration--> 38
1.2354498195337755e+73 1.743325908730129e+71 39
m-->-9.2361093978545e+74, b-->-1.3032944401757952e+73, iteration--> 39
-9.2361093978545e+74 -1.3032944401757952e+73 40
m-->6.904830569430028e+76, b-->9.743309551513593e+74, iteration--> 40
6.904830569430028e+76 9.743309551513593e+74 41
m-->-5.1619879257396465e+78, b-->-7.284008746619922e+76, iteration--> 41
-5.1619879257396465e+78 -7.284008746619922e+76 42
m-->3.8590547700696805e+80, b-->5.4454580489637934e+78, iteration--> 42
3.8590547700696805e+80 5.4454580489637934e+78 43
m-->-2.8849939078971557e+82, b-->-4.0709744310486707e+80, iteration--> 43
-2.8849939078971557e+82 -4.0709744310486707e+80 44
m-->2.1567949522658408e+84, b-->3.043423100358961e+82, iteration--> 44
2.1567949522658408e+84 3.043423100358961e+82 45
m-->-1.6124001001825472e+86, b-->-2.2752351616741986e+84, iteration--> 45
-1.6124001001825472e+86 -2.2752351616741986e+84 46
m-->1.205415508014523e+88, b-->1.7009449130842325e+86, iteration--> 46
1.205415508014523e+88 1.7009449130842325e+86 47
m-->-9.011575643026856e+89, b-->-1.2716107970211733e+88, iteration--> 47
-9.011575643026856e+89 -1.2716107970211733e+88 48
m-->6.73697119624385e+91, b-->9.50644554484022e+89, iteration--> 48
6.73697119624385e+91 9.50644554484022e+89 49
m-->-5.0364977998203355e+93, b-->-7.106931390384198e+91, iteration--> 49
-5.0364977998203355e+93 -7.106931390384198e+91 50
m-->3.7652395043247156e+95, b-->5.313076643566596e+93, iteration--> 50
3.7652395043247156e+95 5.313076643566596e+93 51
m-->-2.8148584767441297e+97, b-->-3.9720073080495614e+95, iteration--> 51
-2.8148584767441297e+97 -3.9720073080495614e+95 52
m-->2.1043623479987168e+99, b-->2.9694361880329186e+97, iteration--> 52
2.1043623479987168e+99 2.9694361880329186e+97 53
m-->-1.573201966727938e+101, b-->-2.219923225450785e+99, iteration--> 53
-1.573201966727938e+101 -2.219923225450785e+99 54
m-->1.1761113434054662e+103, b-->1.6595942175003844e+101, iteration--> 54
1.1761113434054662e+103 1.6595942175003844e+101 55
m-->-8.792500399449483e+104, b-->-1.2406973967315583e+103, iteration--> 55
-8.792500399449483e+104 -1.2406973967315583e+103 56
m-->6.573192555942148e+106, b-->9.275339803093222e+104, iteration--> 56
6.573192555942148e+106 9.275339803093222e+104 57
m-->-4.914058392331553e+108, b-->-6.934158860128532e+106, iteration--> 57
-4.914058392331553e+108 -6.934158860128532e+106 58
```

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m-->3.673704927663267e+110, b-->5.183913486540299e+108, iteration--> 58
3.673704927663267e+110 5.183913486540299e+108 59
m-->-2.746428067806074e+112, b-->-3.875446117978944e+110, iteration--> 59
-2.746428067806074e+112 -3.875446117978944e+110 60
m-->2.053204402682062e+114, b-->2.8972479290702195e+112, iteration--> 60
2.053204402682062e+114 2.8972479290702195e+112 61
m-->-1.5349567565993393e+116, b-->-2.1659559459645366e+114, iteration--> 6
1
-1.5349567565993393e+116 -2.1659559459645366e+114 62
m-->1.147519575524115e+118, b-->1.619248774945083e+116, iteration--> 62
1.147519575524115e+118 1.619248774945083e+116 63
m-->-8.578750968388107e+119, b-->-1.2105355143747146e+118, iteration--> 63
-8.578750968388107e+119 -1.2105355143747146e+118 64
m-->6.413395444169771e+121, b-->9.049852340398738e+119, iteration--> 64
6.413395444169771e+121 9.049852340398738e+119 65
m-->-4.794595539008396e+123, b-->-6.765586503699128e+121, iteration--> 65
-4.794595539008396e+123 -6.765586503699128e+121 66
m-->3.584395595561952e+125, b-->5.0578903408957744e+123, iteration--> 66
3.584395595561952e+125 5.0578903408957744e+123 67
m-->-2.6796612312664607e+127, b-->-3.7812323715822e+125, iteration--> 67
-2.6796612312664607e+127 -3.7812323715822e+125 68
m-->2.0032901288136784e+129, b-->2.826814597441225e+127, iteration--> 68
2.0032901288136784e+129 2.826814597441225e+127 69
m-->-1.4976413038246702e+131, b-->-2.113300634037238e+129, iteration--> 69
-1.4976413038246702e+131 -2.113300634037238e+129 70
m-->1.1196228857024775e+133, b-->1.579884147288881e+131, iteration--> 70
1.1196228857024775e+133 1.579884147288881e+131 71
m-->-8.370197877071222e+134, b-->-1.1811068802294852e+133, iteration--> 71
-8.370197877071222e+134 -1.1811068802294852e+133 72
m-->6.257483068271705e+136, b-->8.82984657399914e+134, iteration--> 72
6.257483068271705e+136 8.82984657399914e+134 73
m-->-4.678036878550832e+138, b-->-6.601112213080647e+136, iteration--> 73
-4.678036878550832e+138 -6.601112213080647e+136 74
m-->3.497257411377369e+140, b-->4.9349308716184166e+138, iteration--> 74
3.497257411377369e+140 4.9349308716184166e+138 75
m-->-2.614517524971459e+142, b-->-3.6893090015034073e+140, iteration--> 75
-2.614517524971459e+142 -3.6893090015034073e+140 76
m-->1.9545892922107482e+144, b-->2.758093530114702e+142, iteration--> 76
1.9545892922107482e+144 2.758093530114702e+142 77
m-->-1.4612330056065003e+146, b-->-2.0619253951785196e+144, iteration--> 7
7
-1.4612330056065003e+146 -2.0619253951785196e+144 78
m-->1.0924043762967593e+148, b-->1.5414764905036712e+146, iteration--> 78
1.0924043762967593e+148 1.5414764905036712e+146 79
m-->-8.1667148002655475e+149, b-->-1.1523936687194201e+148, iteration--> 7
9
-8.1667148002655475e+149 -1.1523936687194201e+148 80
m-->6.105360988663608e+151, b-->8.615189241521826e+149, iteration--> 80
6.105360988663608e+151 8.615189241521826e+149 81
m-->-4.564311808792862e+153, b-->-6.44063636254709e+151, iteration--> 81
-4.564311808792862e+153 -6.44063636254709e+151 82
m-->3.4122375935785656e+155, b-->4.8149605994303476e+153, iteration--> 82
3.4122375935785656e+155 4.8149605994303476e+153 83
m-->-2.55095748993323e+157, b-->-3.599620327718377e+155, iteration--> 83
Yes is close
-2.55095748993323e+157 -3.599620327718377e+155 84
m-->1.9070723936963203e+159, b-->2.6910431012158882e+157, iteration--> 84
Yes is close
1.9070723936963203e+159 2.6910431012158882e+157 85
m-->-1.4257098086310357e+161, b-->-2.0117991102666633e+159, iteration--> 8
5
```

```
Yes is close
-1.4257098086310357e+161 -2.0117991102666633e+159 86
m-->1.0658475604520867e+163, b-->1.504002540219827e+161, iteration--> 86
Yes is close
1.0658475604520867e+163 1.504002540219827e+161 87
m-->-7.968178483758064e+164, b-->-1.1243784876154271e+163, iteration--> 87
Yes is close
-7.968178483758064e+164 -1.1243784876154271e+163 88
m-->5.956937061627691e+166, b-->8.405750320259259e+164, iteration--> 88
Yes is close
5.956937061627691e+166 8.405750320259259e+164 89
m-->-4.4533514439372305e+168, b-->-6.284061748316326e+166, iteration--> 89
Yes is close
-4.4533514439372305e+168 -6.284061748316326e+166 90
m-->3.3292846437761036e+170, b-->4.69790685567668e+168, iteration--> 90
Yes is close
3.3292846437761036e+170 4.69790685567668e+168 91
m-->-2.4889426264287455e+172, b-->-3.512112023808041e+170, iteration--> 91
Yes is close
-2.4889426264287455e+172 -3.512112023808041e+170 92
m-->1.8607106512310068e+174, b-->2.6256226968127724e+172, iteration--> 92
Yes is close
1.8607106512310068e+174 2.6256226968127724e+172 93
m-->-1.391050195709939e+176, b-->-1.9628914166990624e+174, iteration--> 93
Yes is close
-1.391050195709939e+176 -1.9628914166990624e+174 94
m-->1.0399363521160535e+178, b-->1.467439597634464e+176, iteration--> 94
Yes is close
1.0399363521160535e+178 1.467439597634464e+176 95
m-->-7.774468669698184e+179, b-->-1.097044367500915e+178, iteration--> 95
Yes is close
-7.774468669698184e+179 -1.097044367500915e+178 96
m-->5.812121383499195e+181, b-->8.20140294841133e+179, iteration--> 96
Yes is close
5.812121383499195e+181 8.20140294841133e+179 97
m-->-4.345088572829764e+183, b-->-6.131293529671573e+181, iteration--> 97
Yes is close
-4.345088572829764e+183 -6.131293529671573e+181 98
m-->3.2483483155283293e+185, b-->4.583698738308483e+183, iteration--> 98
Yes is close
3.2483483155283293e+185 4.583698738308483e+183 99
m-->-2.4284353706796445e+187, b-->-3.426731084051723e+185, iteration--> 99
Yes is close
```

```
C:\Users\Varun Ladha\Anaconda3\lib\site-packages\ipykernel_launcher.py:12: RuntimeWarning: overflow encountered in double_scalars
  if sys.path[0] == '':
```

In []: